Optimal Impedance Force-Tracking Control Design With Impact Formulation for Interaction Tasks

Loris Roveda, Niccolò Iannacci, Federico Vicentini, Nicola Pedrocchi, Francesco Braghin, and Lorenzo Molinari Tosatti

***Abstract*—The letter presents a force-tracking impedance controller granting a free-overshoots contact force (mandatory performance for many critical interaction tasks such as polishing) for partially unknown interacting environments (such as leather or hard-fragile materials). As in many applications, the robot has to gently approach the target environment (whose position is usually not well-known), then execute the interaction task. Therefore, the algorithm has been designed to deal with both the free space approaching motion (phase a.) and the succeeding contact task (phase b.) without switching from different control logics. Control gains have to be properly calculated for each phase in order to achieve the target force tracking performance (i.e., free-overshoots contact force). In detail, phase a. control gains are optimized based on the impact collision model to minimize the force error during the following contact task, while phase b. control gains are analytically calculated based on the solution of the LQR optimal control problem. The analytical solution grants the continuous adaptation of the control gains during the contact phase on the estimated value of the environment stiffness (obtained through an on-line extended Kalman filter). A probing task has been carried out to validate the performance of the control with partially unknown contact environment properties. Results show the avoidance of force overshoots and instabilities.**

***Index Terms*—Compliance and impedance control, force control, optimization and optimal control, industrial robots.**

I. Introduction

ROBOTS interacting with the environment require a fine control of the interaction, needed to avoid, or at least to limit, the force overshoots during the task execution. Since the milestones of sensor-based force/dynamics control, impedance control and admittance control [1], including also non-restrictive assumptions [2] on the dynamical properties of the environment, have been preferred for robotized assembly tasks. In fact, despite it is demonstrated that position-based force controllers are equivalent to impedance controllers [3], pure force control schemes are better working when some compliance in the force/torque sensor or in the robot joints [4], [5] is present. Impedance control, instead, compounds an easier tunable dynamic balance response for the robot when it executes assembly task. In addition, particular design of impedance controllers [6], grants a wide control bandwidth, thanks to a continuous adaptation of the controller. However, impedance (and admittance) controllers do not allow, practically, a fine control of the interaction force, introducing the problem of not controlled force overshooting during the task execution, and especially at the contact transients. To overcome this limitation, preserving the properties of impedance control, two different families of methods have been mainly introduced: (a) set-point deformation and (b) variable impedance adaptation.

For class (a), the most straightforward solution is suggested in [7], where the time-varying controlled force is derived from a position control law, scaling the trajectory as a function of the estimated environment stiffness. Another important approach is in [8], it involves the generation of a reference motion as a function of the force-tracking error, under the condition that the environment stiffness is variously unknown, *i.e.*, estimated as a function of the measured force. Commonly in (a), all approaches maintain a constant dynamic behavior of the controlled robot, so that when the environment stiffness quickly changes, the bandwidth of the controllers has to be limited for avoiding instability.

Class (b) methods introduce the modification of the dynamic behavior (*i.e.*, online modification of the impedance parameters) during the task execution. Common solutions consist on gain–scheduling strategies that select the stiffness and damping parameters from a predefined set (off-line calculated) on the basis of the current target state [9]–[11]. Such approaches are used in tasks characterized by a stationary, known and structured environment. Similar techniques have been developed also for partially unknown and time-varying environments in applications characterized by a low mechanical bandwidth [12]–[14].

Despite the force overshoots control is of primary importance in many industrial tasks (*e.g.*, fragile components assembly, polishing), at the best of authors' knowledge few works deal with this issue. In [15], we investigated the possibility to adopt a class (b) controller to guarantee the force overshoots avoidance. In that work, the impedance control parameters (*i.e.*, stiffness and damping) were adapted based on the force tracking error. Although the experimental validation shows the capabilities of the defined controller to avoid force overshoots, we were not able to analytically calculate the control gains, having a time-variant closed-loop system (stiffness and damping are functions

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Fig. 1. Experimental Set-up. One KUKA LWR4+ is used as variable stiffness environment. The second KUKA LWR4+ implements the optimal impedance force-tracking controller described in the letter.

of time). In addition, in [16], [17] the authors have shown how the estimation of the environment parameters may improve the performance. Based on these considerations, the here presented letter proposes a class (a) controller, with a two-layer structure.

The internal control layer consists of a pure impedance control with constant parameters. The external control layer consists of an admittance controller providing adaptive control gains online calculated on the basis of the (continuously) updated environment stiffness estimation. The admittance regulator gains are, indeed, analytically calculated as solution of the Linear Quadratic Regulator (LQR) problem. The adoption of Optimal Control methods, indeed, seems suitable to allow the design of controller with large bandwidth. [18], for example, proposes an optimal force controller for an elastic robot moving in contact with the surrounding environment.

The environment stiffness is online estimated through an Extended Kalman Filter as in [19]. In fact, as in common industrial applications, the real stiffness of the interacting environment is not well-know and only the material informations are available. Therefore, the EKF can be initialized based on the target interacting material properties. With respect to the previous work [20], authors are also taking into account the free space approaching motion to the target environment. The same controller - with different control gains - is used also in the free space approaching motion. A formulation of the impact collision is, indeed, introduced for the gain computation. System stability during the control gains transition just after the impact collision is achieved using a logistic function. The proposed control logic is validated in a probing task using a KUKA LWR4+ in contact with a second impedance controlled KUKA LWR4+ that simulates different interacting environments (Fig. 1).

II. Environment Dynamics

A. Compliant Environment Dynamics

Denoting $\mathbf{D}_e$ and $\mathbf{K}_e$ as the environment damping and stiffness respectively, environment dynamics results in [21]:

$$\mathbf{f}_e = -\left(\mathbf{D}_e \dot{\mathbf{x}}_e + \mathbf{K}_e\, \Delta\mathbf{x}_e\right), \quad \text{with} \quad \Delta\mathbf{x}_e := \mathbf{x}_e - \mathbf{x}_e^0 \quad (1)$$

where, $\mathbf{x}_e$ is the environment current position, $\mathbf{x}_e^0$ is the environment equilibrium position and $\mathbf{f}_e$ is the external force/torque. It must be highlighted that the environment mass $\mathbf{M}_e$ is not taken into account. In fact, only local deformations of the interacting environment are considered (as in many works, such as in [8]).

B. Environment Observer: Extended Kalman Filter Design

The environment model in (1) is used to implement an Extended Kalman Filter for the environment stiffness estimation. The filter dynamics is defined by [16]:

$$\boldsymbol{f}\left(\boldsymbol{\xi}_a, \boldsymbol{\nu}_e\right) = \begin{bmatrix} \mathbf{D}_e^{-1}\left(-\mathbf{K}_e \Delta\mathbf{x}_e + \mathbf{f}_e + \boldsymbol{\nu}_{\mathbf{x}_e}\right) \\ \boldsymbol{\nu}_{\mathbf{K}_e} \\ \boldsymbol{\nu}_{\mathbf{D}_e} \\ \boldsymbol{\nu}_{\mathbf{f}_e} \end{bmatrix}$$

where the vector $\boldsymbol{\xi}_a = [\Delta\mathbf{x}_e, \mathbf{K}_e, \mathbf{D}_e, \mathbf{f}_e]^T$ is the augmentated state and the vector $\boldsymbol{\nu}_e = [\boldsymbol{\nu}_{\mathbf{x}_e}, \boldsymbol{\nu}_{\mathbf{K}_e}, \boldsymbol{\nu}_{\mathbf{D}_e}, \boldsymbol{\nu}_{\mathbf{f}_e}]^T$ accounts for uncertainties in models parameters/estimates.

Denoting $\mathbf{y} = [\mathbf{x}_e, \mathbf{f}_e]^T$ as the measurements vector, the observer of the augmented state is therefore defined as:

$$\begin{cases} \dot{\hat{\boldsymbol{\xi}}}_a = \mathbf{f}\left(\hat{\boldsymbol{\xi}}_a, \boldsymbol{\nu}_e\right) + \mathbf{K}_{EKF}(\mathbf{y} - \mathbf{C}_a \hat{\boldsymbol{\xi}}_a) \\ \hat{\mathbf{y}} = \mathbf{H}\left(\hat{\boldsymbol{\xi}}_a, \mathbf{w}\right) \end{cases}$$

where $\hat{\boldsymbol{\xi}}_a$ are estimates, $\mathbf{K}_{EKF}$ is the gain matrix, $\mathbf{C}_a$ is the observation matrix, $\mathbf{H}\left(\boldsymbol{\xi}_a, \mathbf{w}\right)$ is the observation function and $\mathbf{w}$ is the measurements noise. More details are in [16].

Remark 1: The environment damping cannot be estimated with sufficient accuracy in assembly tasks (target task of the present work) since the environment dynamics is not sufficiently excited, and the standard sinusoidal excitation methods [22] cannot be used.

III. Methodology

The control problem of a stable force tracking along a task direction, preserving the impedance behavior along other directions, has the objective of limiting/eliminating any force overshoot. The problem is therefore formulated in term of Optimal Control [23] for the established contact with the environment (Section IV-A), then extended to the *foregoing* transient from free motion to contact (Section IV-B). The algorithm uses the estimate of the stiffness environment (as described in Section II-B) in order to on-line adapt the control gains to optimize the target interaction.

A. Internal Loop: Impedance Control Loop

The goal design for the internal loop consists on achieving a pure decoupled second-order impedance behavior for the controlled robot up to a reasonable frequency around $5\ Hz$. Such behaviour can be achieved by properly design a control loop around the standard position controller for many lightweight industrial robots. Therefore, the target dynamics for the controlled robot should result in:

$$\mathbf{M}_r \ddot{\mathbf{x}}_r + \mathbf{D}_r \dot{\mathbf{x}}_r + \mathbf{K}_r \boldsymbol{\Delta}\mathbf{x}_r = \mathbf{f}_r, \quad \text{with} \quad \Delta\mathbf{x}_r := \mathbf{x}_r - \mathbf{x}_r^0 \quad (2)$$

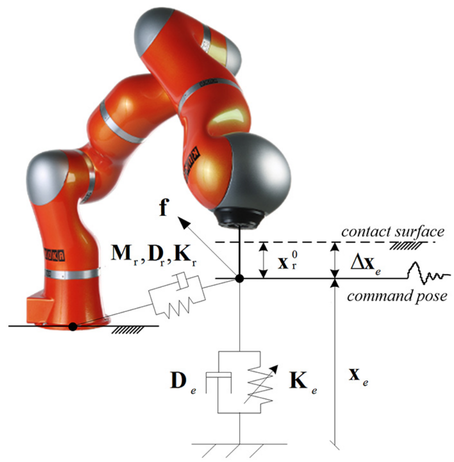

Fig. 2. KUKA LWR4+ interaction model.

where $\mathbf{x}_r^0$ and $\mathbf{x}_r$ are the desired and actual robot poisitions respectively, and $\mathbf{f}_r$ is the external interacting force/torque (Fig. 2). In addition, the Cartesian stiffness $\mathbf{K}_r$, damping $\mathbf{D}_r$ and mass $\mathbf{M}_r$ have to present negligible/ininfluent extra-diagonal coupling terms with a good approximation up to few Hz. In particular, $\mathbf{D}_r = 2\,\mathbf{h}\,\mathbf{M}_r\,\boldsymbol{\omega}_0$, where $\mathbf{h}$ is diagonal matrix of the imposed adimensional damping and $\boldsymbol{\omega}_0 = \sqrt{\mathbf{M}_r^{-1}\mathbf{K}_r}$ is the system pulsation.

Finally, considering a stable contact point with $\mathbf{x}_e^0 = 0$, the environment position is equal to the robot position (*i.e.*, $\mathbf{x}_e = \mathbf{x}_r$) and the force acting on the environment is equal to the force acting on the robot (*i.e.*, $\mathbf{f}_e = \mathbf{f}_r = \mathbf{f}$), as in Fig. 2.

Remarkably, the KUKA LWR4+, that is the robot used for the experimental tests, already displays such behavior of [24]. In fact, experiments show that also $\mathbf{M}_r$ presents negligible/ininfluent extra-diagonal coupling terms with a good approximation up to $5\ Hz$ [25].

B. External Loop: Admittance Control Loop

Considering the interaction dynamics of the system as in (1)-(2), the problem can be formulated as *1 DoF*-task without loss of generality. Therefore, the control design will be treated considering scalar notation instead of matrix notation.

In order to design an optimal control impedance algorithm able to prevent overshooting in contact condition, a first-order reference model for the force response has been selected. In such a way, the target interaction force can be shaped according to the closed-loop dynamics of the coupled system. Thus, denoting f_m as the model reference force, we can introduce the following relation:

$$\dot{f}_m = -c\,f_m + c\,f_d \tag{3}$$

where f_d is a desired constant force, and the parameter c affects the settling time (*i.e.*, the bandwidth) required to the closed-loop system.

The proposed external control scheme, depicted in Fig. 3, is

$$x_r^0 = \ x_r + F_1\,\dot{x}_r + F_2\,x_r + F_3\int_0^t e\,dt + F_4\,f_m.$$

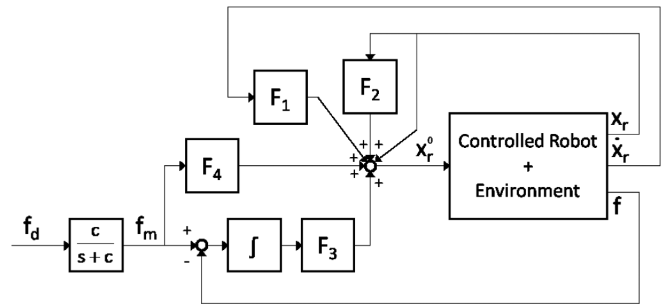

Fig. 3. Optimal control scheme.

The control loop displays four regulators: F_1 is the gain on the robot feedback velocity $\dot{x}_r$, F_2 is the gain on the robot feedback position x_r, F_3 is the gain of the integral of the force tracking error $e_f = f_m - f$, and F_4 is the gain on the model reference force f_m feedforward. Therefore, considering (1), (2), and (3), the corresponding state expression of the coupled robot-environment dynamics results in:

$$\begin{bmatrix}\dddot{x}_r\\ \ddot{x}_r\\ \dot{e}_f\\ \dot{f}_m\end{bmatrix} = \begin{bmatrix} -2h\omega_0-\frac{D_e}{M_r} & -\omega_0^2-\frac{K_e}{M_r} & 0 & 0\\ 1 & 0 & 0 & 0\\ D_e & K_e & 0 & 1\\ 0 & 0 & 0 & -c\end{bmatrix}\begin{bmatrix}\ddot{x}_r\\ \dot{x}_r\\ e_f\\ f_m\end{bmatrix} + \begin{bmatrix}\omega_0^2\\ 0\\ 0\\ 0\end{bmatrix}\dot{x}_r^0,$$

that can be easily rewritten in matrix form as:

$$\dot{\boldsymbol{\eta}} = \mathbf{A}\boldsymbol{\eta} + \mathbf{b}\,\dot{x}_r^0. \tag{4}$$

It is worth to underline that, the environmental parameters in (4) has to be correctly estimated in order to guarantee the desired behavior. The validation of the EKF has been provided in the article of the authors' [16] and in Section V.

Remark 2: If the environment mass M_e has to be taken into account, (4) results in:

$$\begin{bmatrix}\dddot{x}_r\\ \ddot{x}_r\\ \dot{e}_f\\ \dot{f}_m\end{bmatrix} = \begin{bmatrix} \frac{-D_r-D_e}{M_{eq}} & \frac{-K_r-K_e}{M_{eq}} & 0 & 0\\ 1 & 0 & 0 & 0\\ D_e & K_e & 0 & 1\\ 0 & 0 & 0 & -c\end{bmatrix}\begin{bmatrix}\ddot{x}_r\\ \dot{x}_r\\ e_f\\ f_m\end{bmatrix} + \begin{bmatrix}\frac{-K_r}{M_{eq}}\\ 0\\ 0\\ 0\end{bmatrix}\dot{x}_r^0$$

where $M_{eq} := M_r + M_e$.

IV. Optimal Gains Calculation

A. Contact Condition With the Environment

For the purpose of optimal control [23], shall J be the controller performance index:

$$J := \int_0^\infty \left(\boldsymbol{\eta}^T\mathbf{Q}\boldsymbol{\eta} + \left(\dot{x}_r^0\right)^2\right)d\tau, \quad \text{with} \quad \mathbf{Q} := diag\,(0,0,q,0) \tag{5}$$

so that the state is optimized with respect to the force error, *i.e.*, $\boldsymbol{\eta}^T\mathbf{Q}\boldsymbol{\eta} = q\,e_f^2$ for the 1D problem in Equation (4). The optimal controller with the action $\dot{x}_r^0$ is obtained by solving the minimization problem:

$$J^*(\boldsymbol{\eta}) := \min_{\dot{x}_r^0}\{J(\boldsymbol{\eta})\}.$$

that, under the assumptions of the Linear Quadratic Regulator (LQR) problem [23], corresponds to solve the corresponding Riccati matrix equation:

$$\mathbf{0} = \mathbf{S}\ \mathbf{A} + \mathbf{A}^T\mathbf{S} + \mathbf{Q} - \mathbf{S}\ \mathbf{b}\ \mathbf{b}^T\mathbf{S} \tag{6}$$

where $\mathbf{S} = \mathbf{S}^T \geq 0$ is the solution symmetric and positive semi-definite constant 4x4 matrix so that $\dot{x}_r^0 = \dot{x}_r - \mathbf{b}^T\mathbf{S}\boldsymbol{\eta}$.

Worthly, (6) is here solved algebraically (see Appendix for the detiled solution VI) granting $\mathbf{S}$ to be updated continuously with respect to the on-line estimated K_e and D_e.

Upon solution of $\mathbf{S}$ (see Equation (17)–(22), the optimal control solution for $\dot{x}_r^0$ is obtained as:

$$\dot{x}_r^0 = \dot{x}_r - \mathbf{b}^T\mathbf{S}\boldsymbol{\eta}\ =\ \dot{x}_r + F_1\ \ddot{x}_r + F_2\ \dot{x}_r + F_3\ e_f + F_4\ \dot{f}_m, \tag{7}$$

$$F_1 = \omega_0^2\ s_{11},\ F_2 = \omega_0^2\ s_{12},\ F_3 = \omega_0^2\ s_{13},\ F_4 = \omega_0^2\ s_{14} \tag{8}$$

that returns the control action:

$$x_r^0 =\ x_r + F_1\ \dot{x}_r + F_2\ x_r + F_3 \int_0^t e\ dt + F_4\ f_m. \tag{9}$$

Remark 3: The optimal control has been imposed on $\dot{x}_r^0$ and not in x_r^0 to have a linear error force term e_f.

Remark 4: Recalling that q is a suitable variable defined by the user, considering (8) and the analytical solution for $\mathbf{S}$ (in Appendix), it is worth to underline that:

- $F_1,\ F_2$ are function of $K_e,\ D_e$ and q;
- F_3 depends only on q, ensuring a static precision with respect to f_d;
- F_4 is function of $K_e,\ D_e$ and does not depend on q.

B. Free Space Motion Approach

The force control law in (7) ensures the optimality of the control action when the manipulator is initially in contact with the environment. However, the contact detection is always characterized by an instantaneous force increase due to the impact. Therefore, the relation between the impact force and the robot velocity is needed to properly set the control gains in the free space approaching motion.

1) Impact Model: following [26], the impact occurs in a small time Δt during wich all velocities remain finite. Thus, there is no change in controlled robot position. Therefore, integrating both sides of (2) in Δt from the impact time t_s:

$$\lim_{\Delta t\to 0}\int_{t_s}^{t_s+\Delta t} M_r\ddot{x}_r dt + \lim_{\Delta t\to 0}\int_{t_s}^{t_s+\Delta t}\left(D_r\dot{x}_r + K_r\Delta x_r\right)dt = \lim_{\Delta t\to 0}\int_{t_s}^{t_s+\Delta t} f dt$$

As $\Delta t \to 0$ the magnitude of the acceleration variation is $\gg$ than the magnitude of the position and velocity variation, the second term of the left side vanishes [26].

Denoting $Imp = \lim_{\Delta t\to 0}\int_{t_s}^{t_s+\Delta t} f dt$ as the impulse at the collision point, the impact equation that defines the relation between the Imp and the instantaneous modification of the robot cartesian velocity, $\Delta\dot{x}_r$, results in:

$$\Delta\dot{x}_r := \left(\dot{x}_r(t_s+\Delta t) - \dot{x}_r(t_s)\right) = M_r^{-1} Imp. \tag{10}$$

Consider now the instantaneous collision between two bodies, where v_1, v_2 are their velocities before collision, $\Delta v_1, \Delta v_2$ are the changing in the velocities of the bodies after collision, and λ is the coefficient of restitution ($0 \leq \lambda \leq 1$). Thus, following [26],

$$(v_1 + \Delta v_1) + (v_2 + \Delta v_2) = -\lambda\,(v_1 + v_2)$$

Under the assumption that the target interacing environment is not moving (*i.e.*, $v_1 = 0,\ \Delta v_1 = 0$) and the manipulator has to approach to it (*i.e.*, $v_2 = \dot{x}_r,\ \Delta v_2 = \Delta\dot{x}_r$), that is a common scenario in many industrial tasks, the dynamics of the collision can be expressed by [26]:

$$(\dot{x}_r + \Delta\dot{x}_r) = -\lambda\dot{x}_r, \tag{11}$$

Finally, considering (10) and (11), the relation between the impact impulse Imp generated by the collision and the robot cartesian velocity $\dot{x}_r$ results in:

$$Imp = -(1+\lambda)M_r^{-1}\dot{x}_r \tag{12}$$

2) Control Gains Calculus: based on the fact that the controller has not any knowledge of the environment position and the integral of the force error will increase indefinitely until collision is detected, a suitable assumption consists on imposing that the control gains F_2 and F_3 are equal to zero, (7). Thus, the control action (9) results in:

$$x_r^0 =\ x_r + F_1\ \dot{x}_r + F_4\ f_d. \tag{13}$$

where f_m has been replaced by f_d under the assumption that the robot is in the steady condition $\dot{f}_m = 0$.

The computation of control gains F_1 and F_4 can be optimized exploiting the impact model (12), despite only one between F_1 and F_4 can be analytically calculated without adding other assumptions.

Computation of F_4***:*** Considering the manipulator moving in the free space at desired constant velocity, $\dot{x}_r \approx \dot{x}_{r,d}$. Thus, the system dynamic (2) results in:

$$0 = D_r\dot{x}_r + K_r\Delta x_r = 2\ h\ \omega_0\ M_r\ \dot{x}_{r,d} + K_r\left(x_r - x_r^0\right).$$

Thus, taking into account the control action (13), F_4 gain results in:

$$F_4 = \frac{1}{f_d}\ \left(\frac{2h\omega_0\ M_r}{K_r}\dot{x}_{r,d} - F_1\right)$$

In such a way, F_4 deforms the impedance control set-point to obtain a target constant velocity in the free space motion.

Computation of F_1***:*** a suitable criteria consists on considering the force tracking during the contact phase. In particular, F_1 can be numerically calculated introducing an optimization criteria taking into account the impulse Imp. In fact, in order to optimize the force tracking after the impact, there is the need to optimize the controlled robot dynamics in the free space motion and to optimize the robot approaching velocity to the target environment.

Among some candidate criteria, we suggest to calculate both F_1^* and $\dot{x}_{r,d}^*$ as the values that minimize the force tracking error $f_m - f$ starting from the contact establishment:

$$\{F_1^*, \dot{x}_{r,d}^*\}: \quad \min_{\dot{x}_{r,d},\, F_1} \left(\int_{t_s}^{\infty} e_f^2 \, dt \right). \tag{14}$$

Such definition allows to calculate the control action x_r^0 granting the best approaching velocity $\dot{x}_{r,d}^*$.

A table of the parameters F_1^* and $\dot{x}_{r,d}^*$ to be used by the controller can be defined based on the system properties and easily offline computed.

Remark 5: A numerical solution of such problem is obtained as a function of the environment K_e and desired force f_d, setting $D_e = 0$ to simulate the worst contact conditions. Since none information about the environment is available before the contact, the K_e value is set based on the knowledge of the target environment material.

C. Transient Impact Condition: Switching Control Gains Strategy From Free Space to Contact

In order to switch from the control gains defined in Sections IV-A, and IV-B during the impact collision, we propose to use a logistic function. Such continuous and differentiable function guarantees, indeed, a smooth control action during the control gains switching. Therefore, the i-th control gain F_i during the impact collision results in:

$$F_i^{impact}(t) = \frac{F_i^{contact} - F_i^{free-space}}{1 + \exp^{-t/\tau_i}} + F_i^{free-space} \tag{15}$$

where τ_i is the time constant of the control gain adaptation. In particular, two time constants are adopted for feedforward, τ_4, and feedback τ_1, τ_2, and τ_3 terms, in order to have a different time-scale for such control actions changes (1 decade of difference).

Remark 6: Dynamics of the logistic functions is faster than dynamics of the observer (at least 1 decade). Therefore, the analytical control gains adaptation defined in Section IV-A will not be affected by the logistic function dynamics.

V. Experimental Results

All the quantities are refered to the robot base reference frame. The impedance loop rate is 200 Hz, synchronously with the environment estimation. Signals are updated to the main LWR4+ control loop, together with the sampling of force (used in the control loop and by the EKF) and kinematics state. The remote controller is a real-time Linux Xenomai PC with RTNet.

The coupled environment is implemented using a second KUKA LWR4+ (see Fig. 1), setting $K_e = K_r^{robot2}$, obviously *without* injecting known parameters K_e into the controller instead of observations. Two level of K_e have been used ($K_{e,1} = 500 N/m$; $K_{e,2} = 20000 N/m$). During all the experimental tests, the controlled (primary) robot settings are $K_r = 2500\ N/m$ and $h = 0.9$, in the contact direction.

Parameters q in Equation (5) and c in Equation (3) have been selected experimentally to extend the bandwidth of the coupled system as much as possible, avoiding any force overshoot. The main limitation in increasing such values is the presence of the low-pass filter on force measurements. Such filter is implemented in the force measurements of the KUKA LWR 4+ and it has a 5 Hz bandwidth.

Fig. 4 shows the obtained force responses in the contact phase with a soft environment ($K_{e,1}$) varying the approaching velocity, while Fig. 5 shows the obtained force responses in the contact phase with a hard environment ($K_{e,2}$) varying the approaching velocity. In theses experiments the EKF is initialized with $\widehat{K}_{e,i^{th},init} = K_{e,i^{th}}/4$.

Both Figures show the influence of the impact velocity on the target environment. In particular, imposing the best approaching velocity $\dot{x}_{r,d}^*$ the controller is capable to track properly the force reference, even with the imposed value for $\widehat{K}_{e,i^{th},init}$, while imposing a different approaching velocity force error increases. As expected, a hard environment is more sensitive approaching velocity variation. In particular, Fig. 5 shows a force overshoot imposing an approaching velocity higher than the best approaching velocity.

Fig. 6 shows the obtained force responses in the contact phase with a soft environment ($K_{e,1}$) varying the environment stiffness initial value $\widehat{K}_{e,1,init}$ of the EKF. Two values are used: $\widehat{K}_{e,1,init} = K_{e,1}/4$ (black line) and $\widehat{K}_{e,1,init} = K_{e,1} \cdot 4$ (gray line). Low environment stiffness value results in higher calculated approaching velocity (higher initial interaction force increment), while high environment stiffness value results in lower calculated approaching velocity (lower initial interaction force increment). The controller is capable to avoid force overshoot if the error in the estimate of the environment stiffness initial value is in an acceptable range.

Fig. 7 shows the obtained estimated of the interacting environment stiffness $\widehat{K}_e$ considering a soft interacting environemnt ($K_{e,1}$). The environment stiffness initial value $\widehat{K}_{e,1,init}$ used in the EKF is set equal to $K_{e,1}/4$. Results show the convergence of the estimation to the target value. The obtained error is related to measurements noise and to uncertainties in the defined contact model. However, the obtained accuracy is sufficient for the aim of the work. Moreover, the dynamics of the estimation is at least one decade faster than the controller dynamics.

VI. Conclusions

In this letter a two layers controller has been described. The set-point of the internal impedance control layer is calculated by an external admittance control layer. An analytical solution of the LQR optimal control problem is proposed based on the estimation of the interacting environment parameters. The same controller has been used in the free space approaching motion to the contact surface. The gains optimization is based on the impact collision model to minimize the force error during the following contact phase. Experimental results show the force overshoots avoidance adapting the control gains to the environment properties, while minimizing the integral of the force error. Results show the capabilities of the developed controller

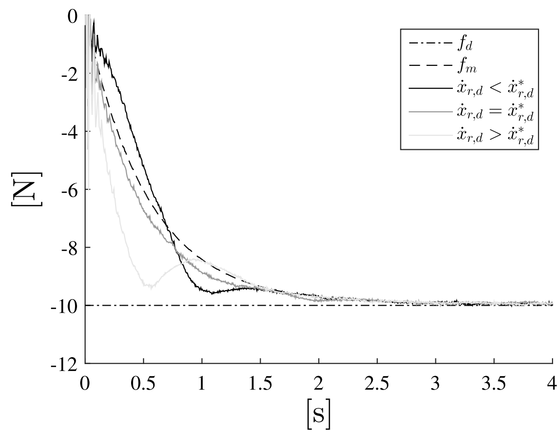

Fig. 4. Comparison between force responses obtained varying the approaching velocity with a soft interacting environment with parameters $q = 1 \cdot 10^{(-5)}$, $c = 2$.

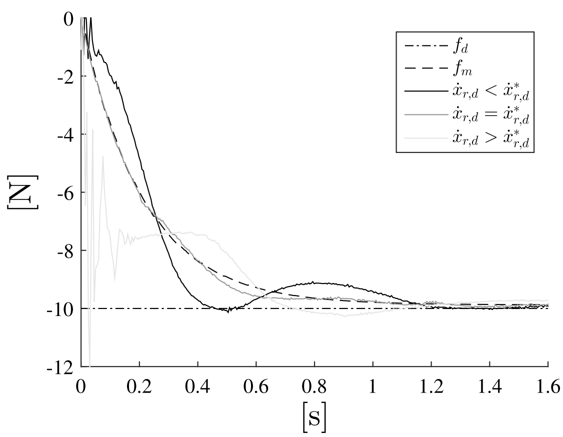

Fig. 5. Comparison between force responses obtained varying the approaching velocity with a hard interacting environment with parameters $q = 5 \cdot 10^{(-5)}$, $c = 5$.

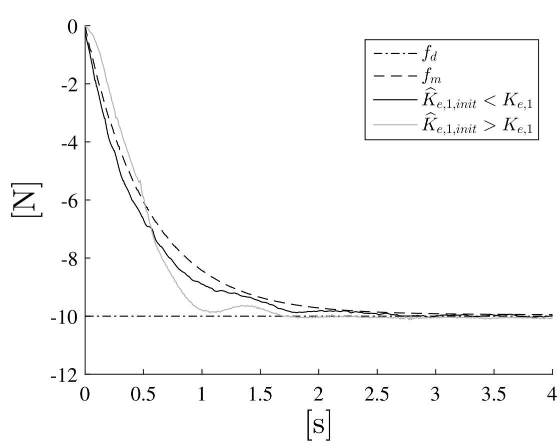

Fig. 6. Comparison between force responses obtained varying the environment stiffness initial value $\widehat{K}_{e,1,init}$ of the EKF with a soft interacting environment.

to obtain the target performance even with errors in the initial estimate of the environment stiffness. Future work will extend the optimal control strategy to the rotational DoFs and will take into account the robot redundancy to optimize the impact dynamics.

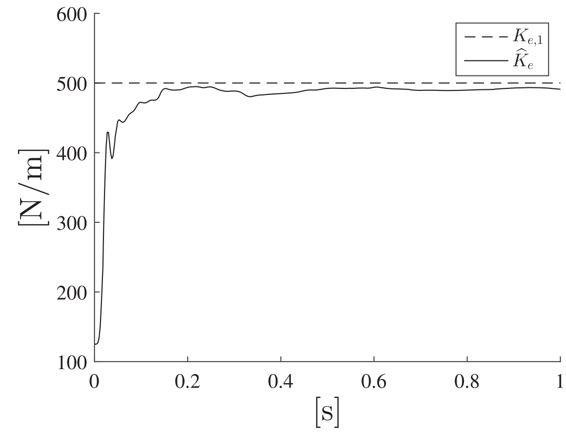

Fig. 7. The estimated stiffness $\widehat{K}_e$ is shown taking into account a soft environment ($K_{e,1}$) and $\widehat{K}_{e,1,init} = K_{e,1}/4$.

APPENDIX

Recalling Equation (6) and denoting s_{nm} the element (n, m) of matrix $\mathbf{S}$, following system of equations gives the solution:

$$0 = 2s_{12} - 2D_e s_{13} - 2s_{11}\left(2h\omega_0 + \frac{D_e}{M_r}\right) - s_{11}^2\omega_0^4 \quad (16a)$$

$$0 = s_{22} - D_e s_{23} - K_e s_{13} - s_{12}\left(2h\omega_0 + \frac{D_e}{M_r}\right) \quad (16b)$$

$$- s_{11}s_{12}\omega_0^4 \frac{K_e s_{11}}{M_r} \quad (16c)$$

$$0 = s_{23} - D_e s_{33} - s_{13}\left(2h\omega_0 + \frac{D_e}{M_r}\right) - s_{11}s_{13}\omega_0^4 \quad (16d)$$

$$0 = -s_{12}^2\omega_0^4 - \frac{2K_e s_{12}}{M_r} - 2K_e s_{23} \quad (16e)$$

$$0 = -s_{12}s_{13}\omega_0^4 - K_e s_{33} - \frac{K_e s_{13}}{M_r} - s_{13}^2\omega_0^4 + q \quad (16f)$$

$$0 = s_{24} - s_{13} - D_e s_{34} - cs_{14} - s_{14}\left(2h\omega_0 + \frac{D_e}{M_r}\right) \quad (16g)$$

$$- s_{11}s_{14}\omega_0^4 \quad (16h)$$

$$0 = -s_{12}s_{14}\omega_0^4 - s_{23} - K_e s_{34} - cs_{24} - \frac{K_e}{M_r}s_{14} \quad (16i)$$

$$0 - s_{13}s_{14}\omega_0^4 - s_{33} - cs_{34} \quad (16j)$$

$$0 = -s_{14}^2\omega_0^4 - 2s_{34} - 2cs_{44} \quad (16k)$$

From (16a), (16c), and (16f) we can obtain the elements

$$s_{12} = \frac{\omega_0^4}{2}s_{11}^2 + \left(2h\omega_0 + \frac{D_e}{M_r}\right)s_{11} - \frac{D_e\sqrt{q}}{\omega_0^2} \quad (17a)$$

$$s_{13} = -\frac{\sqrt{q}}{\omega_0^2} \quad (17b)$$

$$s_{22} = D_e s_{23} + K_e s_{13} + s_{12}\left(2h\omega_0 + \frac{D_e}{M_r}\right) + s_{11}s_{12}\omega_0^4 \quad (17c)$$

$$+ \frac{K_e}{M_r}s_{11} \quad (17d)$$

$$s_{23} = -\frac{s_{12}^2\omega_0^4}{2K_e} - \frac{s_{12}}{M_r} \quad (17e)$$

$$s_{33} = \frac{s_{12}\omega_0^2\sqrt{q}}{K_e} + \frac{\sqrt{q}}{M_r\omega_0^2}. \tag{17f}$$

Substituting Equations (17) into Equation (16d):

$$\begin{aligned} 0 = &\left(\frac{-\omega_0^{12}}{8K_e}\right) s_{11}^4 + \left(-\frac{\omega_0^8\left(2h\omega_0 + \frac{D_e}{M_r}\right)}{2K_e}\right) s_{11}^3 \\ &+ \left(-\frac{\omega_0^4}{2M_r} - \frac{\omega_0^4\left(\left(2h\omega_0 + \frac{D_e}{M_r}\right)^2 - D_e\omega_0^2\sqrt{q}\right)}{2K_e} + \right. \\ &\left. -\frac{D_e\omega_0^6\sqrt{q}}{2K_e}\right) s_{11}^2 + \left(\omega_0^2\sqrt{q} - \frac{2h\omega_0 + \frac{D_e}{M_r}}{M_r}\right) s_{11} \\ &+ \left(\frac{D_e^2 q}{2K_e} + \frac{2h\sqrt{q}}{\omega_0} + \frac{D_e\sqrt{q}}{M_r\omega_0^2}\right), \end{aligned} \tag{18}$$

that can be expressed as the fourth order polynomial:

$$a s_{11}^4 + b s_{11}^3 + c s_{11}^2 + d s_{11} + e = 0 \tag{19}$$

Defining the coefficients:

$$\begin{aligned} &p_0 = \frac{8ac - 3b^2}{8a^2}, p_1 = \frac{b^3 - 4abc + 8a^2 d}{8a^3} \\ &\Delta_0 = c^2 - 3bd + 12ae \\ &\Delta_1 = 2c^3 - 9bcd + 27b^2 e + 27ad^2 - 72ace \\ &\Delta_3 = \sqrt[3]{\frac{\Delta_1 + \sqrt{\Delta_1 - 4\Delta_0^3}}{2}}, \Delta_4 = \frac{1}{2}\sqrt[2]{-\frac{2}{3}p_0 + \frac{1}{3a}\left(\Delta_3 + \frac{\Delta_0}{\Delta_4}\right)} \end{aligned} \tag{20}$$

the polynomial four solutions for s_{11} are:

$$s_{11} = -\frac{b}{4a} - \Delta_4 \pm \frac{1}{2}\sqrt{-4\Delta_4^2 - 2p_0 pm \frac{p_1}{\Delta_4}} \tag{21}$$

s_{11} is selected as solution s.t. it is positive and real, (*i.e.*, $\mathbf{S}$ is positive semi-definite). The remaining elements of $\mathbf{S}$ can be computed from Equations (16j), (16k), (16h) and (16i):

$$s_{44} = -\frac{s_{14}^2\omega_0^4 + 2s_{34}}{2c},\ s_{34} = -\frac{s_{13}s_{14}\omega_0^4 - s_{33}}{2c} \tag{22a}$$

$$s_{24} = s_{13} + c\ s_{14} + s_{14}\left(2h\omega_0 + \frac{D_e}{M_r}\right) + s_{11}s_{14}\omega_0^4 \tag{22b}$$

$$-\ \frac{D_e\left(s_{13}s_{14}\omega_0^4 + s_{33}\right)}{c} \tag{22c}$$

$$s_{14} = \frac{-\left(K_e s_{33}\right)/c + c\ s_{13} - D_e s_{33} + s_{23}}{den} \tag{22d}$$

where

$$\begin{aligned} den = &\left(K_e s_{13}\omega_0^4\right)/c - K_e/M_r - s_{12}\omega_0^4 \\ &- c\left(c + 2h\omega_0 + D_e/M_r + s_{11}\omega_0^4\right) + D_e s_{13}\omega_0^4. \end{aligned} \tag{23}$$

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